Explainable Financial Fraud Detection Using Graph Neural Networks: A Comparative Study

Abstract

Financial fraud detection is difficult because fraudulent activities often involve complex relationships among transactions and financial entities. Traditional machine learning methods usually analyze transactions independently and may miss such relational patterns. Graph Neural Networks (GNNs) are suitable for this problem because they learn from both node features and graph structure. However, GNN predictions are often difficult to interpret, which is a major limitation in financial applications where decisions should be understandable.

This study focuses on explainable financial fraud detection using Graph Neural Networks through a comparative analysis of GNN models. GNNExplainer is used as a post-hoc explainability technique to identify important graph connections and node features responsible for individual predictions. The aim is to compare GNN-based fraud detection models while also examining how explainability can improve understanding of their decisions.

1. Introduction

Financial fraud has become increasingly complex with the rapid growth of digital transactions and online financial services. Traditional fraud detection methods often rely on individual transaction features, but many fraudulent activities involve relationships among multiple entities. Representing financial data as a graph allows such relationships to be analyzed directly.

Graph Neural Networks are designed to learn from graph-structured data by combining the features of a node with information from its neighboring nodes. This makes them suitable for detecting suspicious patterns in transaction networks. Recent studies have shown that GNNs can capture complex relational information that may be difficult to identify using conventional machine learning techniques.

Despite their effectiveness, GNNs are generally difficult to interpret. In fraud detection, simply predicting that a transaction or entity is fraudulent is not always sufficient; it is also important to understand which relationships and features influenced that prediction.

GNNExplainer was introduced to address this problem by identifying a compact subgraph and a subset of important node features that contribute to a GNN prediction.

Therefore, this research investigates **Explainable Financial Fraud Detection Using Graph Neural Networks: A Comparative Study**. The study aims to compare different GNN-based models for fraud detection and use GNNExplainer to interpret their predictions.

2. Literature Review

Graph Neural Networks have gained attention in financial fraud detection because financial systems naturally contain relationships among accounts, transactions, users, and other entities. GNNs can model these relationships and learn representations using both node features and graph connectivity. Existing research shows that GNN-based approaches are useful for detecting complex fraud patterns in financial networks.

A major challenge of GNN-based fraud detection is interpretability. GNN predictions depend on both graph structure and node features, making it difficult to understand why a particular prediction was made.

Ying et al. proposed **GNNExplainer**, a model-agnostic explanation method for Graph Neural Networks. It identifies a small subgraph and a subset of node features that are most influential for a particular prediction. The method learns graph and feature masks and aims to preserve the information most important to the GNN's output.

GNNExplainer is particularly relevant to financial fraud detection because it can help reveal which neighboring transactions or graph connections influenced a fraud prediction. However, it generates explanations individually for each instance, which may become computationally expensive on very large graphs.

Other methods such as PGExplainer, xFraud, and SEFraud have also been proposed to improve efficiency or integrate explainability more directly into fraud detection systems. These approaches will be studied in greater detail in later stages of this work.